

feedWeb — Technology Stack

Learning Management System with AI-powered quiz generation

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Project Layout

```
feedWeb/  
% % % frontend/      -> React SPA          (port 5173 dev)  
% % % backend/       -> Node.js REST API   (port 3001)  
% % % ai-service/    -> Python FastAPI    (port 8000)
```

Frontend talks to the backend; the backend calls the AI microservice via `AI_SERVICE_URL=http://localhost:8000`.

Frontend

Path: frontend/ · Build tool: Vite 5 · Language: JavaScript (JSX)

Core

- **React 18.2** — UI framework
- **Vite 5** — Dev server and bundler
- **Tailwind CSS 4** — Utility-first styling (@tailwindcss/vite)
- **React Router 6** — Client-side routing

Networking & Real-time

- **Axios** — HTTP client to backend REST API
- **Socket.IO Client 4.7** — Real-time events / live updates

UX / Visualisation

- **Framer Motion 12** — Animations / transitions
- **Lucide React** — Icon set
- **Tiptap (ProseMirror)** — Rich-text WYSIWYG editor
- **Chart.js + react-chartjs-2** — Charts and dashboards
- **Recharts 2** — Additional charting
- **react-hot-toast** — Toast notifications

Backend

Path: backend/ · Entry: src/app.js · Runtime: Node.js + Express 4

Web & Real-time

- **Express 4** — REST API framework
- **Socket.IO 4.7** — WebSocket server
- **@socket.io/redis-adapter** — Horizontal scaling for Socket.IO
- **Redis 4.6** — Pub/sub backplane
- **CORS** — Cross-origin support

Database & ORM

- **MySQL** — Relational DB (training_db)
- **mysql2** — Native driver
- **Sequelize 6** — ORM, models, migrations

Auth & Validation

- **JSON Web Tokens (jsonwebtoken)** — Stateless authentication
- **bcryptjs** — Password hashing
- **express-validator** — Request validation

File Uploads & Documents

- **Multer 2** — Multipart upload middleware
- **multer-storage-cloudinary + Cloudinary** — Media CDN storage
- **pdf-parse** — Extract text from PDF uploads
- **Mammoth** — Extract text from DOCX uploads

Email / OTP

- **Nodemailer (Gmail SMTP)** — Transactional email and OTP delivery

AI Integration

- **Axios** — Calls the Python AI microservice
- **OpenAI SDK** — Optional direct LLM integration (currently placeholder key)

Operations

- **Winston** — Structured logging
- **dotenv** — Configuration loading
- **Nodemon, Jest, Supertest, kill-port** — Dev / testing tooling

Layout: src/{routes, controllers, models, middleware, services, socket, jobs, helpers} + database/migrations/.

AI Service

Path: ai-service/ · Entry: main.py · Runtime: Python 3.12 + FastAPI

Web Layer

- **FastAPI 0.115** — REST API: /generate-quiz, /evaluate, /health
- **Uvicorn 0.34** — ASGI server (port 8000)
- **Pydantic 2** — Request/response validation
- **python-multipart** — File-upload support

LLM Stack

- **Google Gemini API** — Primary LLM — gemini-2.5-flash with JSON mode
- **langchain-google-genai** — LangChain integration for Gemini
- **Groq (llama-3.3-70b-versatile)** — Fallback LLM via langchain-openai
- **LangChain core + text-splitters** — Prompt templates, chunked retrieval

Document Parsing & Robustness

- **PyPDF2** — PDF text extraction
- **python-docx** — DOCX text extraction
- **json-repair** — Auto-repair malformed LLM JSON output
- **python-dotenv** — Configuration loading

The quiz generator uses Bloom's Taxonomy-aware prompts, in-memory caching with content-hash keys, and a 3-tier JSON parser (strict -> json-repair -> per-object brace extraction) for resilient output handling.

Infrastructure & Tooling

- **MySQL** — Primary relational database
- **Redis** — Pub/sub for Socket.IO scaling, optional caching
- **Cloudinary** — Media CDN / file storage (CLOUD_NAME=dm9wlkpqc)
- **Gmail SMTP** — Transactional email provider
- **Google Gemini API** — AI quiz generation backend
- **Git** — Version control
- **start-all.bat / start-all.ps1** — Dev orchestration scripts (Windows)
- **start-ai-service.bat** — Standalone AI service launcher

Application Domain

feedWeb is a Learning Management System / Training platform. Workflow files in the repo (QUIZ_WORKFLOW.txt, TRAINER_ADMIN_WORKFLOW.txt, CURRENT_WORKFLOW.txt) describe the following capabilities:

- **AI quiz generation** — Upload course material (PDF/DOCX/TXT) and generate MCQs via Gemini
- **Role-based access** — Trainer / Admin / Trainee
- **Real-time collaboration** — Socket.IO-driven live updates
- **Online proctoring** — frontend/src/proctoring/ module
- **Course materials & feedback** — Materials upload, trainee feedback collection
- **Analytics dashboards** — Chart.js / Recharts visualisations

Security Notes

A few items observed during the tech-stack review that are worth addressing:

- **backend/.env contains live secrets** — DB password, JWT secret, Cloudinary API secret, Gmail app password, Groq API key — all in plaintext.
- **Verify .gitignore** — Ensure backend/.env, ai-service/.env, and frontend/.env are git-ignored.
- **Rotate any shared keys** — API keys that have been pasted in chat or committed to Git should be regenerated.
- **JWT_SECRET placeholder** — The default value contains the literal phrase "change-in-production" — update before any deployment.

None of these are functional bugs — the project runs fine — but they should be tightened before any non-local deployment.